

RS Vorlesung 4

$$HA1) \Rightarrow G_S = \frac{10^{-1}}{s} \cdot \frac{1}{1 + \frac{s}{10^{-1}}} \cdot \frac{1}{1 + \frac{s}{1}}$$

$$HA2) \begin{pmatrix} s^2 + s + 1 & -(s+1) \\ -(s+1) & 2s+1 \end{pmatrix} \begin{pmatrix} \Theta_1 \\ \Theta_2 \end{pmatrix} = \begin{pmatrix} M(s) \\ 0 \end{pmatrix}$$

$$\Theta_2 = \frac{s+1}{2s^3 + 3s^2 + 3s + 1 - s^2 - 2s - 1} M(s)$$

$$\frac{\Theta_2}{M(s)} = \frac{s+1}{2s^3 + 2s^2 + s}$$

$$HA3) \frac{U_a}{U_e} = \frac{R_2 \parallel \frac{1}{sC_2} + R_1 + \frac{1}{sC_1}}{R_1 + \frac{1}{sC_1}} = \frac{\frac{R_2 \cdot \frac{1}{sC_2}}{R_2 + \frac{1}{sC_2}} + R_1 + \frac{1}{sC_1}}{R_1 + \frac{1}{sC_1}}$$

$$= \frac{(sR_2C_2 + 1)sC_1 \left(\frac{R_2}{sR_2C_2 + 1} + R_1 + \frac{1}{sC_1} \right)}{(sR_2C_2 + 1)sC_1 \left(R_2 + \frac{1}{sC_1} \right)}$$

$$= \frac{sC_1R_2 + s^2R_2R_1C_2C_1 + sR_1C_1 + s^2R_2C_2C_1 + 1}{s^2R_2^2C_2C_1 + sR_2C_1 + sR_2C_2 + 1}$$

$$\frac{U_a}{U_e} = \frac{s^2(R_2R_1C_2C_1) + s(R_2C_1 + R_1C_1 + R_2C_2) + 1}{s^2(R_2^2C_2C_1) + s(R_2C_1 + R_2C_2) + 1}$$